Università di Pisa

Mott localisation, antiferromagnetism and nematic superconductivity in the extended Hubbard model with non-local pairing

Candidato:
Alessandro Gori

Relatore:
Prof. Giacomo Mazza

Outline

Cuprates and Hubbard physics

Nobel Prize in Physics 1987

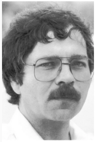


Photo from the Nobel Foundation archive.

J. Georg Bednorz

Prize share: 1/2

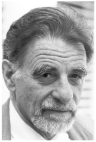


Photo from the Nobel Foundation archive.

K. Alexander Müller

Prize share: 1/2

Possible High T_c Superconductivity in the Ba – La – Cu – O System

J.G. Bednorz and K.A. Müller
IBM Zürich Research Laboratory, Rüschlikon, Switzerland

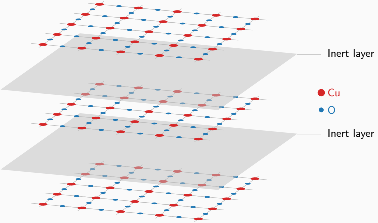
Received April 17, 1986

What makes cuprates interesting?

- Strong "unconventional" insulators;
- Rich phase diagram;
- Origin of Cooper pairing: still unclear;
- Highest critical temperatures for SC transition in this class of materials.

These materials are stackings of:

- 2D copper oxide active layers;
- intermediate layers of inert material.



Most of the cuprates physics is well described by the single-band Hubbard model on the square lattice:

$$\hat{H}_{\mathrm{HM}} = \underbrace{-t \sum_{\langle ij \rangle} \sum_{\sigma} \hat{c}^{\dagger}_{i\sigma} \hat{c}_{j\sigma}}_{\text{Kinetic operator}} \underbrace{+U \sum_{i} \hat{n}_{i\uparrow} \hat{n}_{i\downarrow}}_{\text{On-site repulsion}} \qquad (t, U > 0) \tag{1}$$

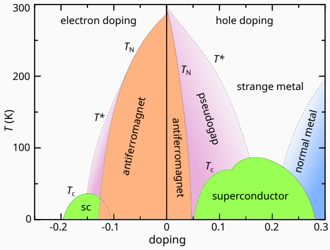
Cuprates exhibit various phases, among which:

- **Antiferromagnetic long-range order** at null and low doping;
- d-**wave superconductivity** as we inject or remove charge.

At **half-filling**, band theory would predict a metallic state. Instead we observe strong insulation.

$\Downarrow$

Cuprates are **not** band insulators.



Schematic phase diagram of cuprates. Author: Holger Motzkau, Wikimedia commons, CC BY-SA 3.0.

Mott localisation: a many-body insulator

Band theory assumption: e-e interactions are **negligible**.

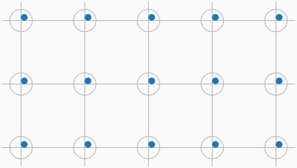
Strongly coupled Hubbard model: on-site repulsion

$$U \sum_i \hat{n}_{i\uparrow} \hat{n}_{i\downarrow}$$

much bigger than kinetics. **Big energetic cost** of having two electrons on the same site.

$$\Downarrow$$

Insulation mechanism: electrons are **localised** as in a sort of “traffic jam”.



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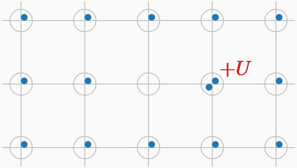
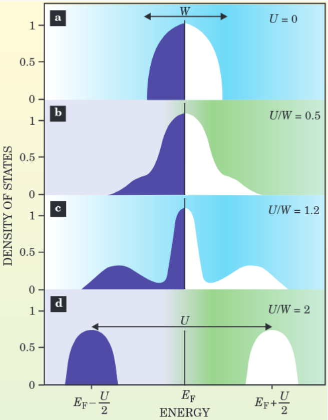


Bands flattening: a signature of Mott localisation



Kotliar, Gabriel, e Dieter Vollhardt, 2004. **Physics Today 57 (March): 53–59**.

(a) Free model, conventional DoS;

(b) Weak coupling, large coherent peak;

(c) Intermediate coupling, **narrow coherent peak**;

(d) Strong coupling, lower and upper Hubbard bands.

$$\Downarrow$$

Localisation = flattening of bare bands.

How does Mott localisation relate to antiferromagnetism and superconductivity?

Antiferromagnetism

Perturbation theory at strong coupling gives:

$$\hat{H}_{\mathrm{HM}} \simeq J \sum_{\langle ij \rangle} \hat{\mathbf{S}}_i \cdot \hat{\mathbf{S}}_j + \Delta E$$

with antiferromagnetic exchange $J = 4t^2/U$.

Mottness: **the ground manifold is localised**.

Superconductivity

Phenomenology: we observe d-wave superconductivity, i.e. the superconducting gap is anisotropic under $\pi/2$ rotations $\mathcal{R}$,

$$\left\langle \hat{c}^{\dagger}_{\uparrow\mathbf{r}} \hat{c}^{\dagger}_{\downarrow\mathbf{r}+\Delta\mathbf{r}} \right\rangle = -\left\langle \hat{c}^{\dagger}_{\uparrow\mathcal{R}\mathbf{r}} \hat{c}^{\dagger}_{\downarrow\mathcal{R}(\mathbf{r}+\Delta\mathbf{r})} \right\rangle$$

Mottness: **non-local Cooper pairing**.

Recent developments: extension of the model with a **non-local attraction**,

$$\hat{H}_{\mathrm{EHM}} = \hat{H}_{\mathrm{HM}} \underbrace{-V \sum_{\langle ij \rangle} \hat{n}_i \hat{n}_j}_{\hat{H}_V}$$

We know from BCS theory that $\hat{H}_V$ can act **as a source of d-wave superconductivity**.

The model (EHM):

$$\hat{H}_{\mathrm{EHM}} = \underbrace{-t\sum_{\langle ij\rangle}\sum_{\sigma}\hat{c}^{\dagger}_{i\sigma}\hat{c}_{j\sigma}}_{\text{Kinetic operator}} \underbrace{+U\sum_{i}\hat{n}_{i\uparrow}\hat{n}_{i\downarrow}}_{\text{Local repulsion}} \underbrace{-V\sum_{\langle ij\rangle}\hat{n}_{i}\hat{n}_{j}}_{\text{Non-local attraction}} \qquad (t, U, V > 0)$$

The motivation: Recent findings of compatibility with this model for experimental results.

The idea: we know that the EHM can host d-wave superconductivity. Can we obtain Mott physics?

The method: theoretical and numerical Hartree-Fock technique.

The Hartree-Fock method